

Shubhangi S. R. Garnaik

shubhangisaille.garnaik@stonybrook.edu • [Portfolio](#) • [LinkedIn](#) • [GitHub](#)

Applied AI researcher and Ph.D. student building human-centered multimodal systems across conversational AI, augmented reality, computer vision, and trustworthy deployment. Experienced in end-to-end prototyping, user evaluation, model testing, and reproducible research software.

Education

Stony Brook University Ph.D. in Computer Science; Advisor: Professor Jihoon Ryoo	2024–Present
Stony Brook University M.S. in Computer Science, GPA: 3.92/4.00	2022–2023

Selected Research Experience

Graduate Researcher, AI2S Lab Stony Brook University	2024–Present
- Developed LocalMate, an accent-adaptive conversational agent combining speech recognition, confidence-aware accent estimation, grounded FAQ retrieval, accent-conditioned speech, and talking-face output; achieved 85% speaker-disjoint and 92.3% out-of-domain accent accuracy. - Developed ArtSpeak, a mobile AR museum application with low-latency semantic retrieval and lifelike portrait responses; delivered responses up to 30× faster than runtime generation, with 85% of study participants reporting useful and relevant interactions.	
Research Intern, MEIC Lab Stony Brook University	Jul.–Dec. 2025
Containerized a reproducible 3D Gaussian Splatting pipeline across preprocessing, training, postprocessing, and rendering; created run scripts, API edge-case tests, and failure-mode documentation.	
Research Assistant, Industry Collaboration Stony Brook University / Taeyang Metal Industrial Co., Ltd.	2025
Built computer-vision workflows for real-time forging monitoring from analog gauges and digital graph displays; analyzed production video constraints and anomaly patterns linked to equipment failures.	

Professional Experience

Data Scientist Thanks Carbon, Republic of Korea	Mar.–Oct. 2024
Developed deep-learning models from weather and rice-yield data to improve agricultural efficiency and support lower-carbon alternate wetting and drying practices.	
Software Test Engineer / Associate Technical Sales Engineer Zensoft / Qualitia, India	2019–2021
Automated software testing and designed structured validation workflows to reduce testing time and improve client software quality.	

Selected Publications and Recognition

LocalMate — Interspeech 2026, accepted • **ArtSpeak** — IEEE ISMAR 2025 • **CLOUD-CODEC** — ACM TOMM 2025
Winner, Decompiler HACKATHON THE X 2026 • **Idea Finalist**, Flipkart Hackathon 2019

Technical Skills

Languages: Python, Java, C#, HTML/CSS **AI/ML:** PyTorch, TensorFlow, OpenCV, WavLM, computer vision, video analytics
Systems: Git/GitHub, Linux, Docker, Jupyter, Unity, Android Studio, 3D Gaussian Splatting **Practice:** API testing, reproducible workflows, technical documentation